

P1979R0: Resolution to [US086](#)

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Effect of this change

This change is intended to prevent `import` declarations in the global module fragment of a module unit from being implicitly imported when other units of the same module import that unit. Other NB comment resolutions will ensure that such `import` declarations must come from `#include` expansion (possibly by translation to an `import`) and must not be explicitly exported.

This is believed to be an unintentional change in behavior from the Modules TS caused by a change in specification mechanisms.

Example

```
// uses_vector.h
import <vector>; // Could be from a translated #include

// partition.cpp
module;
#include "uses_vector.h" // textually expands to import <vector>;
module A:partition;

// interface.cpp
module A;
import :partition;
std::vector<int> x;
```

This change makes the last line ill-formed because it refers to `std::vector` which is not visible. Previously the import of `<vector>` was implicitly imported into the `interface.cpp` translation unit.

Wording

Modify 10.3 [module.import] paragraph 6 as follows:

When a *module-import-declaration* imports a translation unit *T*, it also imports all translation units imported by exported *module-import-declarations* in *T*; such translation units are said to be exported by *T*. **Additionally, when a *module-import-declaration* in a module unit of some module *M* imports another module unit *U* of the same module *M*, it also imports all translation units imported by non-exported *module-import-declarations* in the module unit purview of *U* that module unit.** [Footnote: This is consistent with the rules for visibility of imported names ([basic.scope.namespace]). — end footnote] These rules may in turn lead to the importation of yet more translation units.